MASKINHALLEN · USER MANUAL · VERSION 4.4

SMUTS

LOOP CHOPPER · SORT · REBUILD · DRIFT

Drop in a loop or a whole library. SMUTS chops it, sorts the slices by sound and builds new beats that drag, interlock and drift out of phase.



VST3 · AU · LV2 · STANDALONE
MACOS · WINDOWS · LINUX

MASKINHALLEN

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01 WELCOME

SMUTS is a loop machine. It listens to your loops, cuts them into pieces, sorts the pieces by how they sound, and builds new techno beats from them.

- **The library:** import whole sample folders at once. SMUTS listens to every file and sorts it into drums, percussion, bass, tone and texture, with tempo.
- **SLUMPA KIT** (random kit): four loops that fit together, and a new beat, in one click.
- Four loop slots, six groups (kick, snare, hat, perc, tone, texture), eight patterns with a chain.
- Polyrhythm: every lane has its own length and rate. **ETTAN** brings them all back to the one.
- **GROOVE** changes what plays, live: drift, ratchets, fills, mutations, accents and pump. **DÉJÀ VU** makes the variations come back.
- Twelve beat-synced effects with **KAOS**, two scenes and a crossfader, eight performance buttons.
- 24 recipes, MIDI in and out, and drag-out MIDI.

ABOUT THE SWEDISH LABELS

SMUTS was built in Stockholm, so the controls have Swedish names. The words you will see most:

ON SCREEN	ENGLISH	ON SCREEN	ENGLISH
`Bygg beat`	Build beat	`Ångra`	Undo
`Bibliotek`	Library	`Slumpa kit`	Random kit
`Loopar / Plats`	Loops / Slot	`Ladda`	Load
`Galenskap`	Madness	`Täthet`	Density
`Sväng`	Swing	`Känslighet`	Sensitivity
`Lim / Band`	Glue / Tape	`Ut`	Output
`Kick / Virvel / Hat`	Kick / Snare / Hat	`Perc / Ton / Textur`	Perc / Tone / Texture
`Ettan`	The one (downbeat)	`Déjà vu`	Déjà vu (repeat)
`Fyll / Mutera`	Fill / Mutate	`Knasigt`	Crazy (effects)
`Recept`	Recipe	`Rundgång`	Feedback loop (resample)

THE WINDOW



02 INSTALL

Unzip the file for your system. You get a folder called **SMUTS**. Then run the installer for your system and restart your DAW (or rescan plug-ins). SMUTS shows up as an instrument.

MACOS

1. Double-click `SMUTS-macOS.zip` in Downloads.
2. Open Terminal and run:
   `bash ~/Downloads/SMUTS/installera.command`
3. Quit your DAW completely and start it again.

VST3, AU and a standalone app. macOS 11 or newer, Intel or Apple Silicon. The installer copies the plug-ins and removes macOS's block on downloaded plug-ins.

WINDOWS

1. Right-click `SMUTS-Windows.zip` and choose **Extract all**.
2. Open the `SMUTS` folder and right-click `installera.bat` → **Run as administrator**.
3. If Windows says "Windows protected your PC": **More info** → **Run anyway**.

VST3 (in C:\Program Files\Common Files\VST3) and a standalone .exe. 64-bit Windows 10 or 11.

LINUX

1. Unzip `SMUTS-Linux.zip`.
2. Run `bash ~/Downloads/SMUTS/installera-linux.sh`. It copies the plug-ins to `~/.vst3` and `~/.lv2`.
3. Rescan plug-ins in your DAW (Bitwig, Reaper, Ardour and others).

VST3 and LV2, 64-bit x86 Linux (built on Ubuntu 22.04).

> SWEDISH OR ENGLISH
>
> The machines have Swedish names on the panel. One click on `SV | EN` in the header switches labels, menus and tooltips to English. Your choice is remembered for every MASKINHALLEN machine.

03 UNLOCK WITH YOUR KEY

Your purchase comes with a licence key: four groups of four characters, like `XXXX-XXXX-XXXX-XXXX`. You find it in the Payhip email and on your receipt.

1. Open SMUTS. Until it is unlocked, the window shows **UNLOCK SMUTS**.
2. Paste the whole key in the field and press `UNLOCK`.

3. Done. You only need to do this once per computer: every MASKINHALLEN machine on the computer finds the key. A bundle key unlocks every machine.

The key is not case sensitive, and I/L and O/0 are treated as the same character, so a typo there does not matter. If it says the key does not work, check that the whole key was pasted. You can use your key on your own computers.

THE FREE DEMO

Every machine has a free demo with the same sound engine. Some presets and functions are locked, and the demo has its own name (SMUTS DEMO), so it can sit next to the full version. Buying the full version is what unlocks everything; install it and paste your key.

UPDATES

Updates are free for life. Download the new zip from your Payhip receipt link and run the installer again. Your projects and your key are kept.

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04 YOUR FIRST BEAT IN ONE MINUTE

1. Click `BIBLIOTEK` and then `IMPORTERA MAPP`. Choose a folder with loops. SMUTS reads it in the background.
2. Close the library and click `SLUMPA KIT`. Four loops that fit together are loaded, and a beat is built.
3. Press play in your DAW (or `SPELA` in SMUTS).
4. Open `RECEPT` in the GROOVE panel and pick a recipe. Click `BYGG BEAT` for a new beat with the same loops.
5. Turn `KAOS` up a little. Hold `FILL` before the drop.

Just want one loop? Drag a file or a folder onto one of the four slots. The mouse wheel over a slot steps through the folder while the beat keeps playing.

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05 THE LIBRARY

The library is your own sorted collection. The files stay where they are; SMUTS only keeps a register in `Music/SMUTS/bibliotek.xml`, shared by every SMUTS on the computer.

- **IMPORTERA MAPP** (import folder), or drag a folder onto the library. Every audio file in it and in its subfolders is read. Files already in the library are skipped.
- SMUTS listens to each file: low end, top end, how often it hits, whether it has a pitch, how long it rings, and its tempo. The file name helps too: a file called *bass_124* is a bass loop at 124 BPM.

- The groups: **TRUMMOR** (drums), **PERC**, **BAS** (bass), **TON** (tonal: synths, chords, melodies), **TEXTUR** (pads, noise, atmospheres) and **ÖVRIGT** (other).
- **Click** a sound to listen. It plays on top of the beat, even when the transport is stopped. **TYST** stops it.
- **Double-click** to load it into the slot chosen under **LADDA I PLATS** (load into slot).
- **Right-click** for more: load into any slot, mark as favourite, change the group if SMUTS guessed wrong, show the file, or remove it from the library.
- **RENSA** (clean) removes files that have been moved or deleted.
- **SLUMPA KIT** picks four sounds at your project tempo (or half or double): drums, percussion, bass or tone, and tone or texture. Favourites are picked more often. Then a new beat is built. `Ångra` (undo) brings the old beat back.

06 LOOPS AND GROUPS

Four slots. The waveform shows the pieces coloured by group.

- **< and >** or the mouse wheel step through the folder. Only the sounds from that slot change; the beat keeps playing.
- **PÅ** (on) decides whether the slot is used when a beat is built. Right-click empties it.
- **KÄNSLIGHET** (sensitivity) sets how many hits are found. Release the knob and the loops are sliced again.

GROUP	SOUNDS LIKE	IN THE BEAT
`KICK`	lots of low end, punchy	four on the floor, always
`VIRVEL`	noise with body	2 and 4, or sparse
`HAT`	bright and short	offbeats, denser with density
`PERC`	everything in between	euclidean rhythm in an odd length
`TON`	has a pitch	a short figure in minor pentatonic
`TEXTUR`	long and soft	a layer on the one

If a loop has no piece for a group, the closest-sounding piece is borrowed.

07 THE BEAT

- **Click** an empty step to turn it on. **Alt-click** toggles directly, **shift-click** sets an accent, **right-click** swaps the piece, pitch or reverse.
- Each lane has a **length** and a **rate** (1/16, 1/8, 1/16T, 1/8T, 1/32, 5-tuplets, 7-tuplets, or **TAKT**, which squeezes the whole lane into one bar). Different lengths slide against each other as polyrhythms.

- **Patterns A–H**: click a letter to edit it. What plays switches on the next beat, so it never stutters. **KEDJA** plays a chain like *AAAB*, each pattern for 1, 2, 4 or 8 bars.
- The lane menu (click the lane name): euclidean fill, *euclid from the loop* (the loop's own rhythm made euclidean), rotate, logic against the kick (AND, OR, XOR, NOT, ECHO), piece choice, call and response, fill holes with echoes.
- **LJUDKARTA** (sound map) shows every piece from all four loops as dots: dark to the left, bright to the right. Click a dot to listen and place it on the selected step.
- **DRA MIDI** (drag MIDI): drag the button to a track to get the pattern as MIDI, four bars.

08 THE INSPECTOR

Shows the step and lane you selected.

- **VILLKOR** (condition): 1:2, 3:4, 75/50/25 % chance, FILL, not fill, first. The beat changes over 8–16 bars without you touching anything.
- **RATCHET**: the step hits 2, 3 or 4 times. **STYRKA** (velocity), **TONHÖJD** (pitch, ±2 octaves) and **BAKLÄNGES** (reverse).
- **START I BITEN** (start in the piece) and **SVEP** (sweep), where the start point walks through the piece every cycle.
- **EFFEKTLÅS** (effect lock): an effect locked to a value for just that step.
- Per lane: **FILTER**, **DRIVE** and **EKO** (echo send).

09 GROOVE, ETTAN AND DÉJÀ VU

GROOVE never changes the steps. It decides live what plays. Everything at zero means SMUTS plays the beat exactly as written.

KNOB	DOES
`DRIFT`	lanes rotate one step at a time against the kick
`RATCHET`	hits roll in 2–4 quick parts
`FYLL`	the last bar of every 8 fills up; high up, everything but the kick drops out just before
`MUTERA`	some steps flip on or off, a new state every other bar
`ACCENT`	a euclidean accent layer: weak hits get quieter and shorter
`PUMP`	the kick pushes all other lanes down, like sidechain
`ETTAN`	every 1, 2, 4 or 8 bars all lanes restart on step one, however long or skewed they are

KNOB	DOES
DÉJÀ VU	locks the variations into a loop as long as the ETTAN period (4 bars if ETTAN is off). At 100 % every period sounds exactly the same; lower, a step here and there changes

> SKEWED, BUT IT ALWAYS LANDS
>
> A 7-step hat against a 16-step kick drifts further apart every bar. With ETTAN on every 4th bar it drifts for four bars and then snaps back to the one. Add MUTERA and RATCHET for movement, and DÉJÀ VU at 80–100 % so the movement becomes a groove you recognise instead of noise.

- **SEED**: a new variant of ratchets, mutations and fills. The same seed always sounds the same.
- **RUNDGÅNG** (resample): SMUTS records the next bar of itself, with the effects, and loads it as loop 4. The file is saved in `Music/SMUTS/Rundgang`.
- The rings on the right show the lanes as circles, kick innermost. The text below says when they all meet again.

10 KNASIGT: THE TWELVE EFFECTS

Everything runs in time with your DAW, and everything is off when the knobs are at the bottom (FILTER in the middle).

KNOB	DOES	KNOB	DOES
GATE	chops the volume in a rhythm	FRYS	freezes a grain for the rest of the beat
FILTER	left dark, right thin	STAMMA	stutters the start of the beat
KROSS	bit crush and down-sample	BAKLÄNGES	reverses pieces in a fixed pattern
HACKA	shuffles the sixteenths	TEJP	tape slows down during the beat
GLITCH	tiny grains that get stuck	SCRATCH	the playhead is dragged like a record
TONHÖJD	octave and fifth jumps	DUBB	ping-pong echo on 3/16

KAOS turns the twelve knobs by itself, with a new nudge on every beat. A thin cyan arc around a knob shows where KAOS has pushed it right now.

BUSS: `LIM` (glue) is a compressor and soft clipper that makes pieces from different loops sound like one beat. `BAND` (tape) adds saturation, a little low end around 90 Hz and a rounded top.

11 LIVE AND MIDI

- **SPARA A / SPARA B** store the twelve effect knobs and KAOS as scenes. The crossfader glides between them, and can be automated.
- The eight **performance buttons** work while held and release on the next sixteenth, so it always stays in time: STAMMA, BAKLÄNGES, BROMS, FRYS, GLITCH, HACKA, MÖRKER and DUBB.

MIDI IN	DOES
`48 (C2 in Bitwig)`	FILL, while held
`49-56`	the eight performance buttons, in order
`57`	BYGG BEAT
`CC1 (mod wheel)`	the crossfader

MIDI out: every lane sends a note when it hits: KICK = C1, VIRVEL = C#1, HAT = D1, PERC = D#1, TON = E1, TEXTUR = F1. Let SMUTS drive another synth or drum machine.

12 NEW IN 3.0: LIVING PATTERNS

SMUTS 3.0 is about beats that keep moving without you touching them, and about getting cleaner sounds out of messy loops.

FEATURE	WHAT IT DOES
`RULLEN`	The first GROOVE knob. Just before each new bar, one or more lanes may change a little: a hit moves a 16th, a slice is swapped for a similar-sounding sibling, a ghost note appears or disappears, SMET switches on, or the lane returns to a variant it played earlier. Every change is checked against the same rhythm rules as SPÅN and BETONG, so changes that would sound wrong are thrown away. The kick's four-on-the-floor is never touched. More RULLEN means more lanes and more often, with bigger moves at the end of four-bar phrases.
`KORS`	Cross-patching. Builds a new beat where each group takes its slices from a different loop: kick from one, hat from another, tone from a third.
`LÅS` and dice per lane	A locked lane is left alone by BYGG BEAT, KORS, the dice and RULLEN. The dice on a lane gives that lane alone a new pattern and new slices, following the rhythm rules. GROOVE, KNASIGT and BUSS have dice too. `Ångra` undoes the latest change, whether it was a pattern or knobs.
`DELA FÖRST`	Split first. Every loop is split into low (below 150 Hz), mid and high (above 5.5 kHz) bands before chopping. You get a clean kick from the lows and a clean hat from the highs, even from a dense loop or a full track.

FEATURE	WHAT IT DOES
SMET	Smear, in the inspector. The slice keeps playing through the loop for 2–8 steps instead of being cut at the next hit. It shows as a band under the steps. Great on tone and texture.
32 steps	Lanes can be up to 32 steps long (the *Längre* submenu on the length button). The grid then shows 32 cells.
Grid detection	SMUTS finds the loop's downbeat. If a loop was cut slightly late, it is rotated so the one comes first before chopping.
GROOVE-DNA	The menu next to RECEPT. The same swing and accent profiles as SPÅN, VENTIL and BETONG. The DNA knob sets the amount. The kick never moves.
HÄMTA VENTIL	Fetch VENTIL. Takes the last four bars VENTIL played (through SLINGAN, the shared loop-back) and loads them into slot 4. The file is saved in *Music/Phunkshunz/SMUTS VENTIL*.
GULD	Always-on recording of the last 10 minutes. Select a part of the waveform and drag DRA UT WAV or DRA UT MIDI into Bitwig. GULD sets a bookmark and stores the pattern. ÅTERKALLA brings that pattern back.

13 NEW IN 4.0: FRIKTION

SMUTS 4.0 stops sitting on a stiff grid. Inspired by Romanian and Berlin minimal, the FRIKTION panel lets the kick stay locked as the anchor while everything else is pulled back, leans on each other and stumbles in a controlled way. Everything on zero = the straight grid, exactly like 3.0. Projects saved before 4.0 open with FRIKTION on zero.

CONTROL	WHAT IT DOES
SLÄP PER SPÅR	Drag per lane, 0–45 ms behind the grid: VIRVEL, HAT, PERC, TON and TEXTUR. The kick never drags, it is the anchor.
TRÖGHET	Inertia. Hits in a row drag more and more: +8, +18, +30 ms, as if pulled through syrup, and snap back after a pause. Ratchet rolls drag progressively too.
JITTER	±1–3 ms of organic flutter per hit. It repeats with the seed, so it is alive but not random noise.
INTERLOCK	The lanes listen to each other, in the order KICK, VIRVEL, HAT, PERC, TON, TEXTUR. When a stronger lane hits, the weaker ones duck (1 ms attack, 35–60 ms release) and get the stronger lane's frequency band carved out of them. Above 50 %, a weaker hit landing on the same step as a stronger one moves 1/32 or 1/64 later as a quieter, shorter, darker ghost note. Above 60 %, tails are choked. Four loops sound like one drummer with eight arms.
DWELL	Chance that the needle gets stuck on a 1/32 or 1/64 of the slice, 2–4 times, and that the next hit then rushes to catch up as a quintuplet burst that lands with a smack.

CONTROL	WHAT IT DOES
TURING	Chance that a slice is swapped for a neighbour in the same group, or played reversed, an octave down or from another start point. The kick's downbeat is never touched.
SVANS	Chance that a hit gets a random tail, from dry 35 ms clicks to 350 ms room tails that read on into the loop.
LIM-MASTER	A master filter that morphs LP → BP → HP and self-oscillates at the top of RESO, an envelope (KUVERT) that opens or closes it on the hits, and TEJP: warm, asymmetric tape saturation that lifts the tails and micro details. FILTER fully open, FORM on LP and KUVERT on zero = off.
VÄVEN	The weave. One row per lane: the vertical line is the grid, the dot is where the hit actually landed, the trail shows the last hits. Cyan = the lane is being ducked. D = dwell, B = burst, S = ghost note.
DICE // POCKET	New polymetric lengths (11, 13, 15, 16, never more than one lane on 11), new euclidean hits and new drag that is guaranteed to swing without derailing. Locked lanes are left alone. Ångra undoes both the knobs and the pattern.
LÅS GROOVET	Lock the groove. All randomness (TURING, DWELL, jitter, mutations, ratchets and RULLEN) is frozen into a loop that repeats exactly. Press it when you hear gold.
Start loops	A new SMUTS now sounds right away: four loops it calculates itself (A sub/kick anchor, B woody clicks, C metallic micro percussion, D smoky dub chord and noise), with lanes on 16, 15, 13 and 11 steps. Drop your own loops in and they are replaced.

14 RECIPES

A recipe sets the build knobs, BUSS, GROOVE and KNASIGT, and builds a new beat. The sounds come from your loops, so the same recipe sounds different with every kit.

CATEGORY	FOR
GRUND	clean, straight beats; a good starting point
SVÄNG	swing, ghost notes and rolling hats
HYPNOS	locked, slowly changing loops for long, hypnotic sets
STÖK	glitch, crush, reverse and scratch, still landing on the one
DUBB	echo, freeze and dark space

15 NEW IN 4.2 – 4.4

TWO TABS AND SIX MACROS (4.3)

The window is 1180 × 720 and split in two tabs:

- `BEATET` (THE BEAT): loops, the beat, the inspector, the macros and LIVE.
- `MASKINRUM` (ENGINE ROOM): GROOVE, FRIKTION, KNASIGT, BUSS and GULD.

Six macro knobs each move several controls at once: `SMUTS`, `SLÄP` (drag), `STÖK` (mess), `RYMD` (space), `KROSS` (crush) and `SVEP` (sweep). They sit on top of your settings, so saved knobs are never touched; all macros at zero sounds exactly like before. `SLUMPA MAKRON` randomises them (with undo).

MUTERA, HACKA I TAKT AND RÖRELSE (4.4)

`MUTERA`	Mutate: two or three unlocked tracks get the roll's changes right away. GALENSKAP sets how strong. Undoable.
`HACKA I TAKT`	Chop in time: one slice per 16th note. The nearest hit within a third of a 16th is used, otherwise the cut lands on a zero crossing on the grid. Long loops get a coarser grid.
`RÖRELSE`	Movement: every macro is modulated slowly in time (3–16 bars, phase from the seed, SVEP moves both ways). The arcs around the macros show where they are.

BETTER CHOPPING AND SMARTER SLICES (4.2)

Hits are found in three bands with an adaptive threshold, every slice starts just before the attack on a zero crossing, and the tempo and grid are worked out from the loop (or its file name). Sorting looks at the first 45 ms after each hit with the previous sound subtracted, so a hat on a kick tail is sorted as a hat. BUILD BEAT picks slices weighted by quality; GALENSKAP flattens the weights for wilder choices.

16 TROUBLESHOOTING

SMUTS does not show up	Quit the DAW completely and start it again, or rescan plug-ins. On Mac, run the installer again: it removes macOS's quarantine on the plug-in. On Windows, check that `SMUTS.vst3` is in C:\Program Files\Common Files\VST3.
macOS says the plug-in is damaged	That is macOS's block on downloaded files. Run `bash ~/Downloads/SMUTS/installera.command` again.
Windows protected your PC	Click **More info** → **Run anyway**. The plug-ins are not signed with a paid certificate.
The key does not work	Paste the whole key, including the dashes. A key for one machine only unlocks that machine; a bundle key unlocks all.

No sound	Press play in the DAW: the machines follow the transport. Instruments go on an instrument track, effects on an audio track or bus.

Anything else? Ask through the store, and the person who built the machine answers.

MASKINHALLEN